



FUTURE INTERNS

APISECURITYRISKANALYSIS REPORT

CYBER SECURITY

TASK3_REPORT

Submitted by:

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Repository:

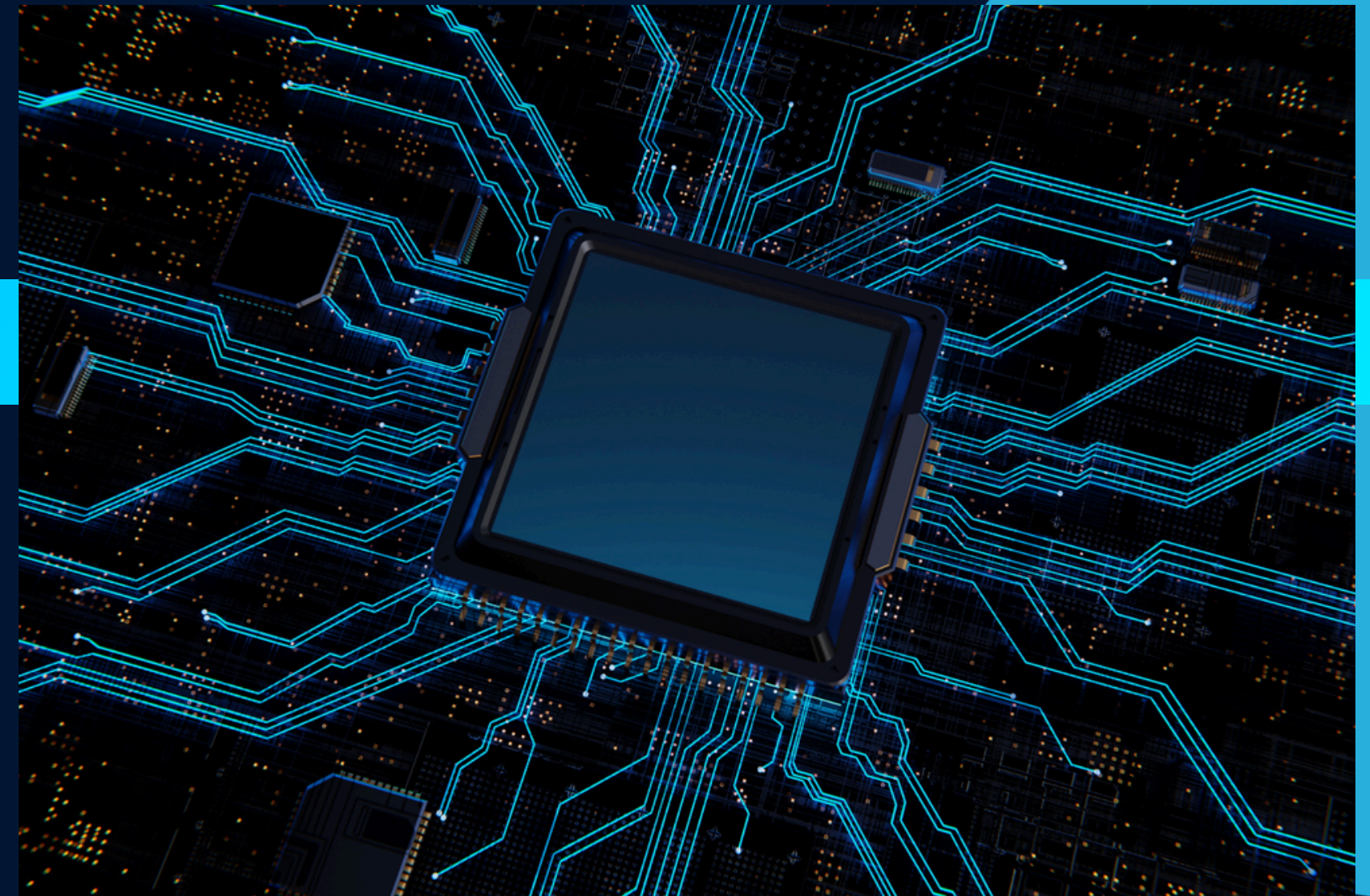
FUTURE_CS_03

INTRODUCTION

This project focuses on analyzing API security risks using Postman and public testing APIs. The objective is to test API endpoints, observe request and response behavior, and identify possible security weaknesses in API communication.

OBJECTIVE

- Test API endpoints using Postman
- Analyze request and response data
- Identify security risks in APIs
- Understand API security concepts





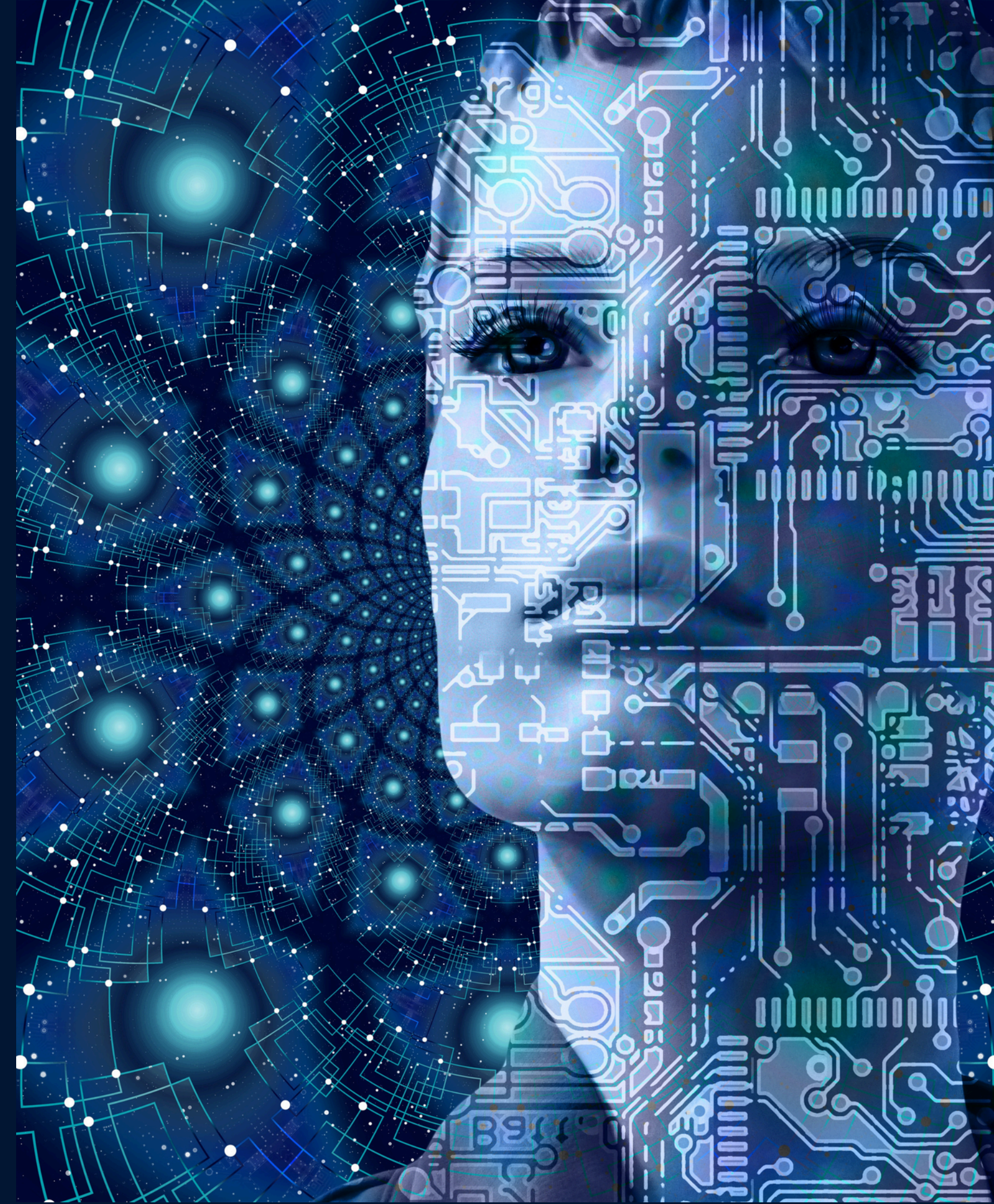
Tools Used

Tool	Purpose
Postman	API Testing
JSONPlaceholder API	Public Testing API



API Endpoints Tested

1. <https://jsonplaceholder.typicode.com/users>
2. <https://jsonplaceholder.typicode.com/posts>





Request & Response Analysis

The API requests were tested using GET methods in Postman. The responses returned JSON formatted data along with HTTP status code 200 OK, indicating successful communication between client and server.



Security Risks

Security Risk	Severity	Observation
Missing Authentication	High	Public access allowed
Excessive Data Exposure	Medium	User data visible
Missing Rate Limiting	Medium	Unlimited requests possible
Public API Endpoint	Medium	Open API accessibility

GET Get dataGET Get dataGET https://jsonplaceholder

https://jsonplaceholder.typicode.com/users

GEThttps://jsonplaceholder.typicode.com/users

DocsParamsAuthorizationHeaders 6BodyScriptsTestsSettings

BodyCookiesHeaders 26Test Results

JSON

PreviewVisualize

1

[

2

{

3

"id": 1,

4

"name": "Leanne Graham",

5

"username": "Bret",

6

"email": "Sincere@april.biz",

7

"address": {

8

"street": "Kulas Light",

9

"suite": "Apt. 556",

10

"city": "Gwenborough",

11

"zipcode": "92998-3874",

12

"geo": {

13

"lat": "-37.3159",

14

"lng": "81.1496"

15

}

16

},

17

"phone": "1-770-736-8031 x56442",

18

"website": "hildegard.org",

19

"company": {

20

"name": "Romaguera-Crona",

21

"catchPhrase": "Multi-layered client-server neural-net",

22

"bs": "harness real-time e-markets"

23

}

24

},

25

{

26

"id": 2,

27

"name": "Ervin Howell",

28

"username": "Antonette",

29

"email": "Shanna@melissa.tv",

30

"address": {

Connect Git

Console

Terminal

0

0

GET Get dataGET Get dataGET https://jsonplaceholder

https://jsonplaceholder.typicode.com/users

GEThttps://jsonplaceholder.typicode.com/users

DocsParamsAuthorizationHeaders 6BodyScriptsTestsSettings

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KeyValue

:status	200
date	Sat, 30 May 2026 15:16:38 GMT
content-type	application/json; charset=utf-8
content-length	1847
access-control-allow-credentials	true
cache-control	max-age=43200
content-encoding	gzip
etag	W/"160d-1eMSsxeJRfnVLRBmYJSbCiJZ1qQ"
expires	-1
nel	{"report_to":"heroku-nel","response_headers":["V"]
pragma	no-cache
report-to	{"group":"heroku-nel","endpoints":[{"url":"https://r
reporting-endpoints	heroku-nel="https://nel.heroku.com/reports?s=P
server	cloudflare
vary	Origin, Accept-Encoding
via	2.0 heroku-router
x-content-type-options	nosniff
x-powered-by	Express

GET Get data

GET Get data

GET https://jsonplaceholder

+

https://jsonplaceholder.typicode.com/posts

GET

https://jsonplaceholder.typicode.com/posts

Docs

Params

Authorization

Headers 6

Body

Scripts

Tests

Settings

Body

Cookies

Headers 24

Test Results

JSON

Preview

Visualize

1

[

2

{

3

"userId": 1,

4

"id": 1,

5

"title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",

6

"body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas ut

7

},

8

{

9

"userId": 1,

10

"id": 2,

11

"title": "qui est esse",

12

"body": "est rerum tempore vitae\nsequi sint nihil reprehenderit dolor beatae ea dolores neque\nfugiat blanditiis voluptate porro quibusdam

13

},

14

{

15

"userId": 1,

16

"id": 3,

17

"title": "ea molestias quasi exercitationem repellat qui ipsa sit aut",

18

"body": "et iusto sed quo iure\nvoluptatem occaecati omnis eligendi aut ad\nvoluptatem doloribus vel accusantium

19

},

20

{

21

"userId": 1,

22

"id": 4,

23

"title": "eum et est occaecati",

24

"body": "ullam et saepe reiciendis voluptatem adipisci\nsit amet autem assumenda provident rerum culpa\nquia hic commodum non facit sed voluptate ipsa quo beatae ut aut qui autem autem delectos re

25

},

26

{

27

"userId": 1,

28

"id": 5,

GET Get data

GET Get data

GET https://jsonplaceholder

+

https://jsonplaceholder.typicode.com/posts

GET

https://jsonplaceholder.typicode.com/posts

Docs

Params

Authorization

Headers 6

Body

Scripts

Tests

Settings

Body

Cookies

Headers 24

Test Results

Key	Value
:status	200
date	Sat, 30 May 2026 15:24:27 GMT
content-type	application/json; charset=utf-8
access-control-allow-credentials	true
cache-control	max-age=43200
content-encoding	gzip
etag	W/"6b80-Ybsq/K6GwwqrYkAsFxqDXGC7DoM"
expires	-1
nel	{"report_to":"heroku-nel","response_headers":["Via"],"max_age":300,"sri":false}
pragma	no-cache
report-to	{"group":"heroku-nel","endpoints":[{"url":"https://nel.heroku.com/reports?r=https://www.jsonplaceholder.typicode.com/&s=6b80-Ybsq/K6GwwqrYkAsFxqDXGC7DoM"}]}
reporting-endpoints	heroku-nel="https://nel.heroku.com/reports?s=WFgQ8Nc2sEastC"
server	cloudflare
vary	Origin, Accept-Encoding
via	2.0 heroku-router
x-content-type-options	nosniff
x-powered-by	Express
x-ratelimit-limit	1000
x-ratelimit-remaining	999

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Terminal

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RECOMMENDATIONS

- Implement API authentication mechanisms
- Apply rate limiting for API requests
- Restrict unnecessary data exposure
- Use proper API security headers
- Monitor API activity regularly

CONCLUSION

This project provided practical exposure to API security testing and risk analysis using Postman. The analysis helped in understanding API request handling, response monitoring, and common API security risks.